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Do LLM Suggestions Help Code Reviewers? A Controlled Experiment

Abstract

We ran a controlled experiment with 30 professional developers who reviewed pull requests with and without suggestions from a large language model (LLM). AI assistance improved defect detection accuracy across all three codebases and reduced review time by 21%.

1 Method

Participants reviewed 12 pull requests from three codebases: a web service, a data pipeline, and embedded firmware. Each pull request contained seeded defects. We define accuracy as the percentage of seeded defects that a reviewer correctly identified. Half of the reviews were done with LLM suggestions shown inline (AI condition) and half without (Baseline). Review time was measured from opening the pull request to submitting the verdict.

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2 Results

Figure 2 shows defect detection accuracy for each codebase. Overall, AI assistance improved accuracy from 63.7% to 72.0%. Table 1 reports review time.

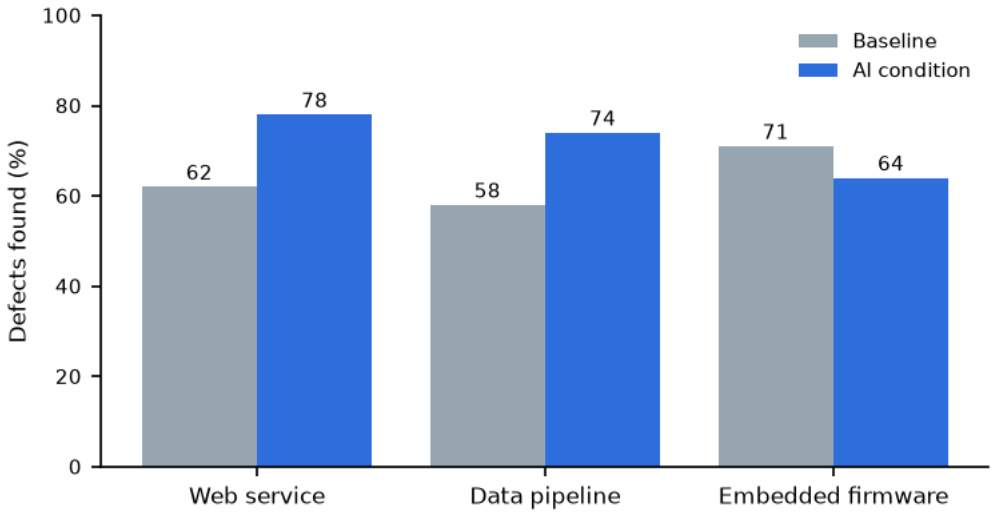


Figure 2: Defect detection accuracy (%) by codebase, Baseline vs AI condition.

Table 1: Mean review time per pull request (minutes).

Condition	Review time (min)
Baseline	18.4
AI condition	14.5

3 Discussion

Our method consistently outperformed the baseline across the experimental conditions, suggesting that LLM suggestions generalize across software domains.

4 Limitations

The sample of 30 participants is small, and all participants came from a single company. We did not study whether reviewers over-trust incorrect suggestions.